T Tutorial L Calculus I.

- function
- limit
- differentiation
- Integration

→ Pre-test - 5%
Tests → 15%
→ 20%
40%
Final - 60%
100%
≥ 50%

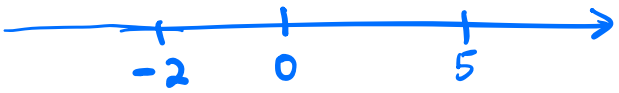
Preliminaries and Functions

Numbers, Absolute Values, and Inequalities

Real Numbers and Real Lines

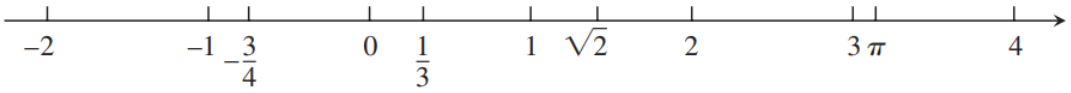
Much of the calculus is based on properties of the real number system. **Real numbers** are numbers that can be expressed as decimals, such as

$-\frac{3}{4} = -0.7500\ldots$
$\frac{1}{3} = 0.33333\ldots..$
$\sqrt{2} = 1.4142\ldots$



Note: The dots "…" in each case indicate the sequence of decimal digits goes forever.

The real numbers can be represented geometrically as points on a number line called **real line**.



We use $\mathbb{R}$ denotes real number system.

-b → inverse of b

The properties of the real number system fall into three categories: *Algebraic Properties, Order Properties, and Completeness.*

a+(-b)

(i) **Algebraic Properties**: the real numbers can be added, subtracted, multiplied, and divided (expect by 0), to produce more real numbers under usual rules of arithmetic.

(ii) **Ordered Properties**:

Rule for Inequality: If a, b and c are real numbers, then

1. $a < b \Rightarrow a + c < b + c$
2. $a < b \Rightarrow a - c < b - c$
3. $a < b$ and $c > 0 \Rightarrow ac < bc$
4. $a < b$ and $c < 0 \Rightarrow bc < ac$
   (Special case: $a < b \Rightarrow -b < -a$)
5. $a > 0 \Rightarrow \frac{1}{a} > 0$
6. If a and b are both positive or both negative, then $a < b \Rightarrow \frac{1}{b} < \frac{1}{a}$.

(iii) **Completeness Property**: Roughly speaking, it says that there are enough real numbers to "complete" the real number line, in the sense that there are no "holes" or "gaps" in it.
Note: The set of rational numbers has all the algebraic and order properties of the real numbers but *lacks the completeness property*.
For example, there is no rational number whose square is 2; there is a "hole" in the rational line where should be. Real numbers that are not rational are called **irrational numbers**.

interval — open $(a,b) \to a < x < b$; closed $[a,b] \to a \le x \le b$; $(a,b] \to a < x \le b$; $[a,b) \to a \le x < b$

Intervals

A subset of real line is called an **interval** if it contains at least two numbers and contains all the real numbers lying between any two of its elements.

For example, if $a < b$, the open interval from a to b consists of all numbers between a and b and is denoted by the symbol (a,b). Using set builder notation, we write

$$(a,b) = \{x \mid a < x < b\}.$$

Notice that the endpoints of the interval, namely, a and b are excluded.

The *closed interval* from a to b consists of all numbers between a and b, including a and b. We denote this closed interval by $[a,b]$. Using set builder notation, we write

$$[a,b] = \{x \mid a \le x \le b\}.$$

$x \ge a$ $[a,\infty)$; $x < a$ $(-\infty, a)$; $(-\infty,\infty) \to x \in \mathbb{R}$ ($\mathbb{R}$ real number)

Intervals can also be half-open (or half-closed), e.g.

$$(a,b] = \{x \mid a < x \le b\}.$$

We also consider infinite intervals such as

$$[a,\infty) = \{x \mid x \ge a\}, \quad (-\infty,b) = \{x \mid x < b\}, \quad (-\infty,\infty) = \{x \mid x \in \mathbf{R}\}.$$

The symbol ∞ or $-\infty$ are not numbers. For example, ∞ simply indicates that the interval extends indefinitely far in the positive direction.

Notation	Set description	Picture
(a, b)	$\{x \mid a < x < b\}$	
$[a, b]$	$\{x \mid a \leqslant x \leqslant b\}$	
$[a, b)$	$\{x \mid a \leqslant x < b\}$	
$(a, b]$	$\{x \mid a < x \leqslant b\}$	
(a, ∞)	$\{x \mid x > a\}$	
$[a, \infty)$	$\{x \mid x \geqslant a\}$	
$(-\infty, b)$	$\{x \mid x < b\}$	
$(-\infty, b]$	$\{x \mid x \leqslant b\}$	
$(-\infty, \infty)$	$\mathbb{R}$ (set of all real numbers)	

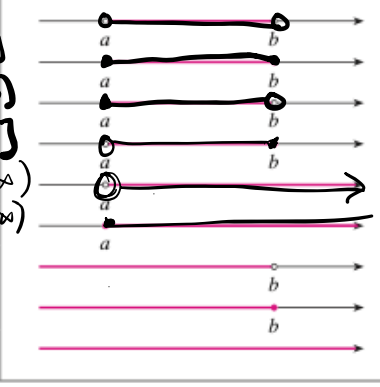


Table of Intervals

Example. Solve the inequality $\dfrac{x+1}{x-1} < 2$.

Solution:

$\dfrac{x+1}{x-1} < 2$ → $x-1 > 0$ or $x-1 < 0$

$\dfrac{x+1}{x-1} - 2 < 0$

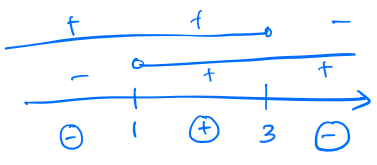
$\dfrac{x+1-2(x-1)}{x-1} < 0$

$\dfrac{-x+3}{x-1} < 0$

$\dfrac{3-x}{x-1} < 0$

$x < 1$, $x > 3$.

let $3-x > 0$, $x < 3$; $x-1 > 0$, $x > 1$



Absolute Value

The *absolute value* of a number a, denoted $|a|$, is the distance from a to 0 on the real line, given a convenient unit of measurement. Distances are always non-negative, so we have

$$|a| \geq 0 \text{ for any number } a.$$

To facilitate algebraic manipulation, we write the absolute value function as a step function:

$$\sqrt{a^2} = |a| = \begin{cases} a & \text{if } a \geq 0, \\ -a & \text{if } a < 0. \end{cases}$$

Let a and b be real numbers, and n be an integer. We have the following properties of absolute values:

1. $|a| = \sqrt{a^2}$.
2. $|ab| = |a||b|$.
3. $\left|\frac{a}{b}\right| = \frac{|a|}{|b|}$, $b \neq 0$.
4. $|a^n| = |a|^n$.
5. Suppose that $a > 0$. Then $|x| < a \Leftrightarrow -a < x < a$.
6. Suppose that $a > 0$. Then $|x| > a \Leftrightarrow x < -a$ or $x > a$.

The Triangle Inequality. If a and b be real numbers, then $|a+b| \leq |a| + |b|$.

Example. Solve the inequality $|3x-2| \leq |x-1|$.

Solution

$$\sqrt{(3x-2)^2} \leq \sqrt{(x-1)^2}$$

$$(3x-2)^2 \leq (x-1)^2$$

$$9x^2 - 12x + 4 \leq x^2 - 2x + 1$$

$$8x^2 - 10x + 3 \leq 0$$

$$(4x-3)(2x-1) \leq 0$$

$$(4x-3)(2x-1) \leq 0$$

Let $4x-3 \geq 0$, $x \geq 3/4$

$2x-1 \geq 0$, $x \geq 1/2$

$$1/2 \leq x \leq 3/4$$

Example. Describe the set of points satisfying the inequality $x^2 + y^2 - 4x - 6y < 12$.

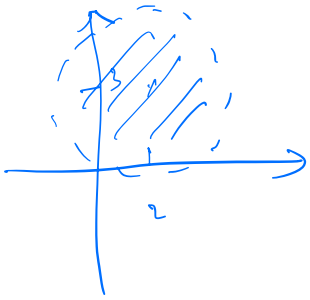
Solution

$(x^2-4x)+(y^2-6y) < 12$.

$(x-2)^2-4+(y-3)^2-9 < 12$

$(x-2)^2+(y-3)^2 < 25 = 5^2$.

Disk.



Extra Examples:

1. Solve the inequality $x^2 - 5x + 6 \leq 0$.

2. Solve $x^3 + 3x^2 > 4x$.

3. Solve the inequality.
   a. $|x + 3| = |2x + 1|$
   b. $|x + 1| \geq 3$
   c. $1 \leq |x| \leq 4$

4. Prove that $|ab| = |a||b|$.

Tues 10-11 am, 11-12 pm BTM4

Thurs 10-11 am, 11-12 pm BTM4

select 1 group.

Quiz - 5% W4

Test 1 - 15%. W7 (online).

Test 2 - 20% W11/W12

Functions and Their Graphs

Functions; Domain and Range (pg 21)

The value of one variable quantity, say y, depends on the value of another variable quantity, which we often call x. Since the value of y is completely determined by the value x, we say that y is a function of x.

Often the value of y is given by a *rule* or formula that says how to calculate it from the variable x.

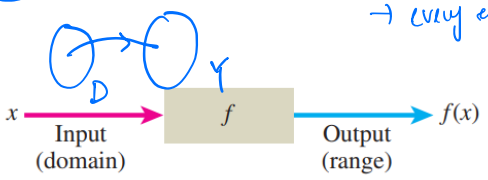
Example: The equation $A = \pi r^2$ is a rule that calculates the area A of a circle from its radius r.

In calculus, we may want to refer to an unspecified function without having any particular formula in mind.
A symbolic way to say **"y is a function of x"** is by writing

$$y = f(x)$$

In this notation, the symbol f represents the function. The letter x, called the **independent variable**, represents the input value of f, and y, the **dependent variable**, represents the corresponding output value of f at x.

Definition:
A **function** f from a set D to a set Y is a rule that assigns a *single (unique)* element $f(x)$ in Y to each $x \in D$.



The set D of all possible input values is called the **domain** of the function.
The set of all output values of $f(x)$ as x varies throughout D is called the **range** of the function.

The range may not include every element in the set Y.

The domain and range of a function can be any sets of objects, but often in calculus they are sets of real numbers interpreted as points of a coordinate line.

Example. Verify the domains and ranges of these functions.

Function	Domain (x)	Range (y)
$y = x^2$	$(-\infty, \infty)$	$[0, \infty)$
$y = 1/x$	$(-\infty, 0) \cup (0, \infty)$	$(-\infty, 0) \cup (0, \infty)$
$y = \sqrt{x}$	$[0, \infty)$	$[0, \infty)$
$y = \sqrt{4 - x}$	$(-\infty, 4]$	$[0, \infty)$
$y = \sqrt{1 - x^2}$	$[-1, 1]$	$[0, 1]$

Example (Exercises 1.1).

Find the domain of $f(x) = 1 - \sqrt{x}$, $g(x) = \sqrt{x^2 - 3x}$ and $G(t) = \frac{2}{t^2-16}$

Solution

$f(x) = 1 - \sqrt{x}$

$D = x \in [0, \infty)$

$g(x) = \sqrt{x^2 - 3x}$

$D = (-\infty, 0] \cup [3, \infty)$

$x^2 - 3x \geq 0$

$x(x-3) \geq 0$

Let $x \geq 0$ $\quad x - 3 \geq 0$

$x \geq 0 \quad x \geq 3$

$x \leq 0, \quad x \geq 3.$

$G(t) = \frac{2}{t^2-16}$

$= \frac{2}{(t-4)(t+4)}$

$t \neq 4, -4$

$D = \mathbb{R} \setminus \{-4, 4\}$

$(-\infty, -4) \cup (-4, 4) \cup (4, \infty)$

Graphs of Function (pg 23)

If f is a function with domain D, its **graph** consists of the points in the *Cartesian plane* whose coordinates are the input-output pairs for f. In set notation, the graph is

$$\{(x, f(x)) \mid x \in D\}.$$

input, output

The graph of a function $f(x) = x + 2$ is the set of points with coordinates (x, y) for which $y = x + 2$.

The graph of a function f is a useful picture of its behavior. If (x, y) is a point on the graph, then $y = f(x)$ is the height of the graph above the point x. The height may be positive or negative, depending on the sign of $f(x)$.

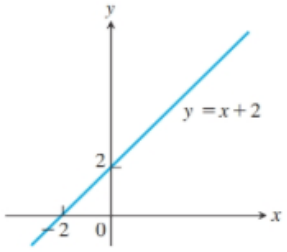


FIGURE 1.3 The graph of $f(x) = x + 2$ is the set of points (x, y) for which y has the value $x + 2$.

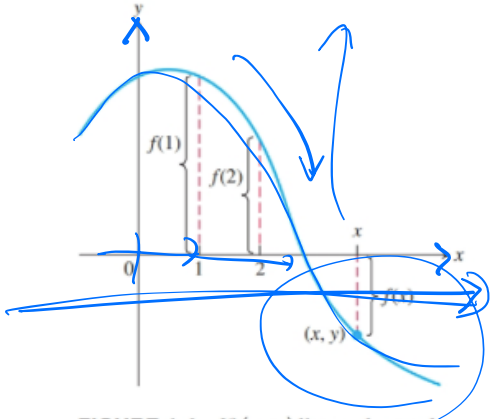


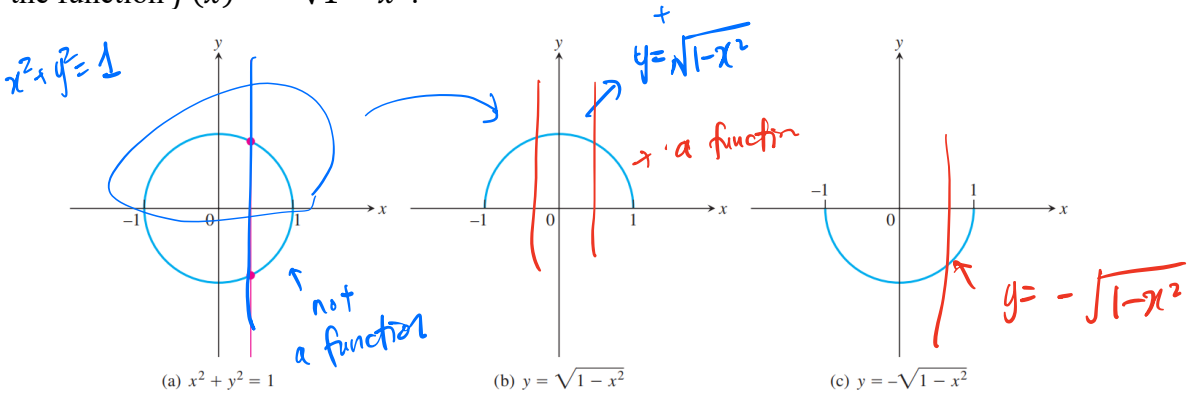
FIGURE 1.4 If (x, y) lies on the graph of f, then the value $y = f(x)$ is the height of the graph above the point x (or below x if $f(x)$ is negative).

The Vertical Line Test for a Function (pg 24)

A function f can have only **one value $f(x)$** for each x in its domain, so *no vertical line* can intersect the graph of a function more than one point. If a is in the domain of a function f, then the vertical line $x = a$ will intersect the graph of f in the single point $(a, f(a))$.

$x^2+y^2=1$

Example. The circle cannot be the graph of a function, since some vertical lines intersect the circle twice. The circle does contain the graphs of two functions of x; namely the upper semicircle defined by the function $f(x) = \sqrt{1-x^2}$ and the lower semicircle defined by the function $f(x) = -\sqrt{1-x^2}$.

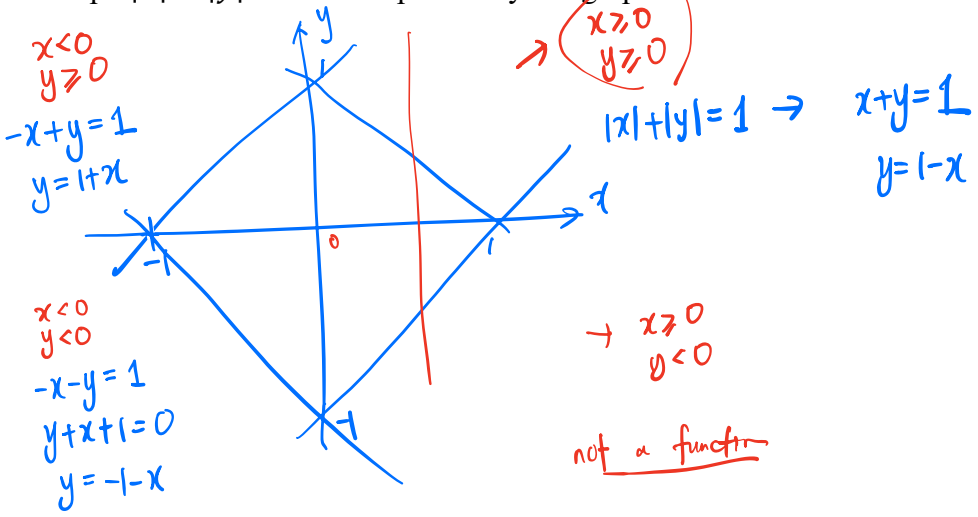


(a) $x^2 + y^2 = 1$ (b) $y = \sqrt{1-x^2}$ (c) $y = -\sqrt{1-x^2}$

$$|x| = \begin{cases} x, & x \geq 0 \\ -x, & x < 0 \end{cases}$$

Examples (Exercises 1.1):

Graph $|x| + |y| = 1$ and explain why not graphs of functions of x.

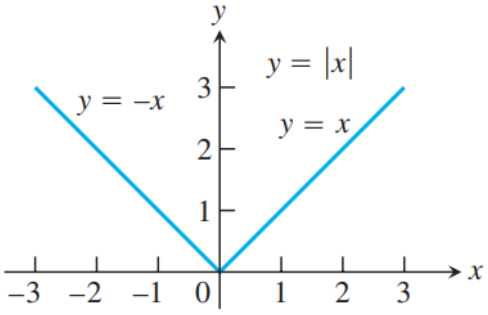


Piecewise-Defined Functions

Sometimes a function is described by using different formulas on different parts of its domain.

Example. Absolute Value Function.

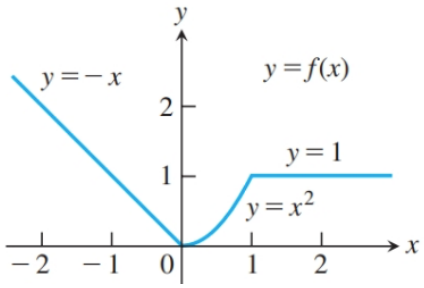
$$|x| = \begin{cases} x, & x \geq 0 \\ -x, & x < 0, \end{cases}$$

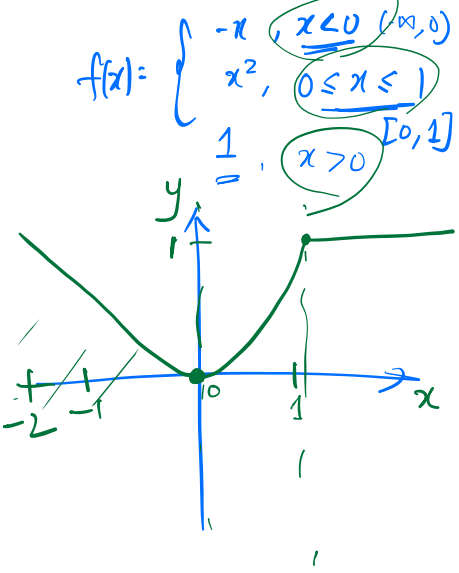


The absolute value function has domain $(-\infty, \infty)$ and range $[0, \infty)$.

Example. The function

$$f(x) = \begin{cases} -x, & x < 0 \\ x^2, & 0 \le x \le 1 \\ 1, & x > 1 \end{cases}$$



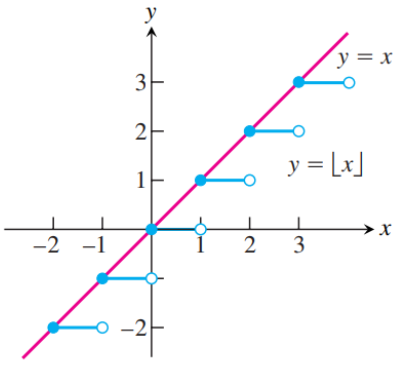


To graph the function $y = f(x)$, we apply different formulas to different parts of its domain.

Example. The Greatest Integer Function (Integer Floor Function)

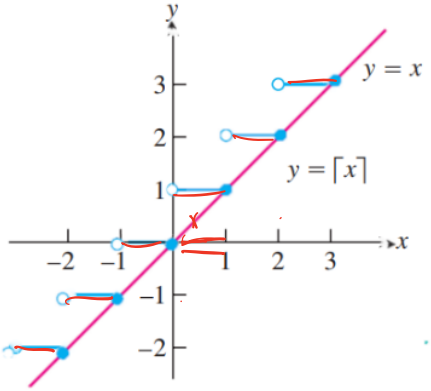
The function whose value at any number x is the *greatest integer less* than or *equal to* x is called the **greatest integer function** or the **integer floor function**. It is denoted $\lfloor x \rfloor$. Observe that

$$\lfloor 2.4 \rfloor = 2, \lfloor 1.9 \rfloor = 1, \lfloor 0 \rfloor = 0, \lfloor 2 \rfloor = 2, \quad \lfloor -1.2 \rfloor = -2, \lfloor -0.3 \rfloor = -1, \lfloor -2 \rfloor = -2.$$



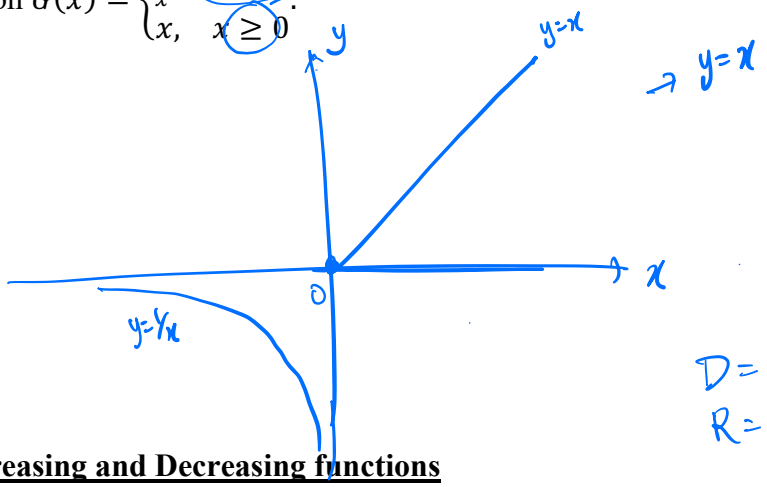
Example. The Least Integer Function (Integer Ceiling Function)

The function whose value at any number x is the *smallest integer greater* than or *equal to* x is called the **least integer function** or the **integer ceiling function**. It is denoted $\lceil x \rceil$.



Examples:

Graph $G(x) = \begin{cases} \frac{1}{x}, & x < 0 \\ x, & x \geq 0 \end{cases}$.



$\rightarrow y = x$

$D = \mathbb{R}$

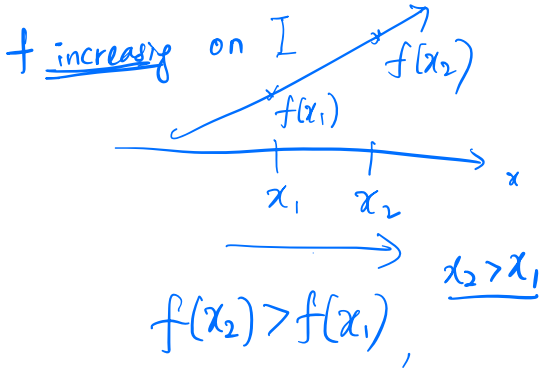
$R = \mathbb{R}$.

Increasing and Decreasing functions

If the graph of a function climbs or rises as you move from left to right, we say that the function is *increasing*. If the graph descents or falls as you move from left to right, the function is *decreasing*.

Definitions Let f be a function defined on an interval I and let x_1 and x_2 be two distinct points in I.

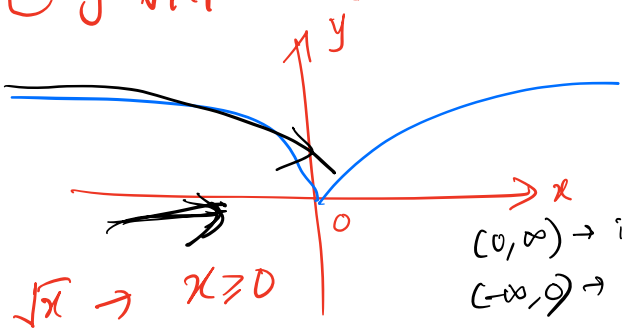
1. If $f(x_2) > f(x_1)$ whenever $x_1 < x_2$, then f is said to be **increasing** on I.
2. If $f(x_2) < f(x_1)$ whenever $x_1 < x_2$, then f is said to be **decreasing** on I.



Examples:

1. $y = \sqrt{|x|}$
2. $y = -4\sqrt{x}$
3. $y = \frac{1}{|x|}$

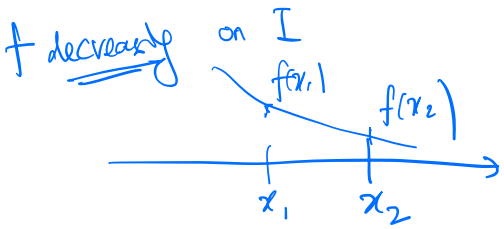
① $y = \sqrt{|x|}$ $D = \mathbb{R}$



$\sqrt{x} \rightarrow x \geq 0$

$(0, \infty) \rightarrow$ increasing

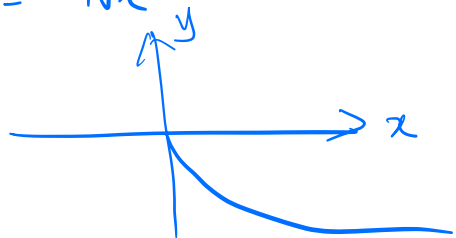
$(-\infty, 0) \rightarrow$ decreasing



$f(x_1) > f(x_2)$

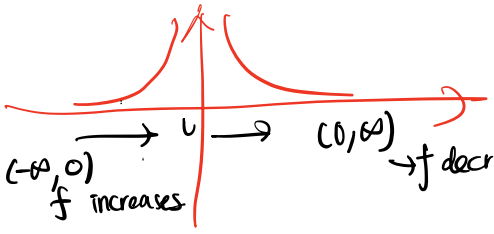
$f(x_2) < f(x_1)$, $x_2 > x_1$

② $y = -4\sqrt{x}$



$(0, \infty)$ decreasing

$y = \frac{1}{|x|} = \left(\frac{1}{x}\right)$



Definitions. Even functions and Odd functions: Symmetry

A function $y = f(x)$ is an

even function of x if $f(-x) = f(x)$,

odd function of x if $f(-x) = -f(x)$,

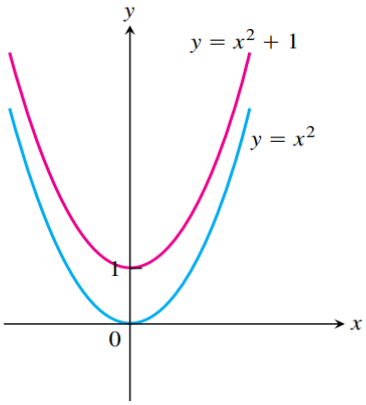
for every x in the function's domain.

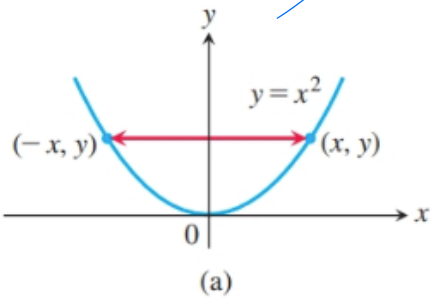
The graph of an *even function* is **symmetric about the y-axis** $(f(-x) = f(x))$.

Note:

1. a point (x, y) lies on the graph if and only if the point $(-x, y)$ lies on the graph.
2. a reflection across the y-axis leaves the graph unchanged.

Example: $y = x^2$, $y = x^2 + 1$.





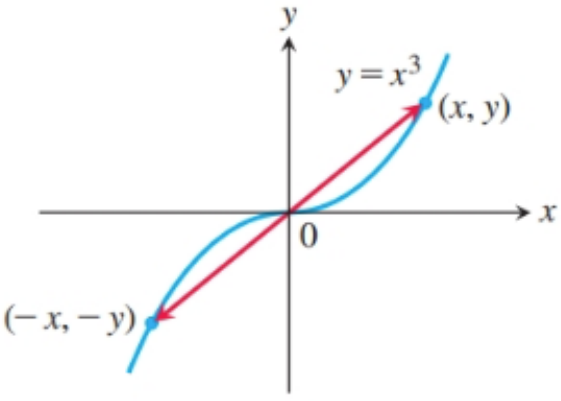
(a)

The graph of an *odd function* is **symmetric about the origin** $(f(-x) = -f(x))$.

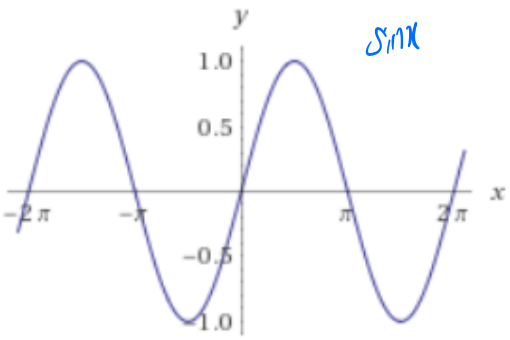
Note:

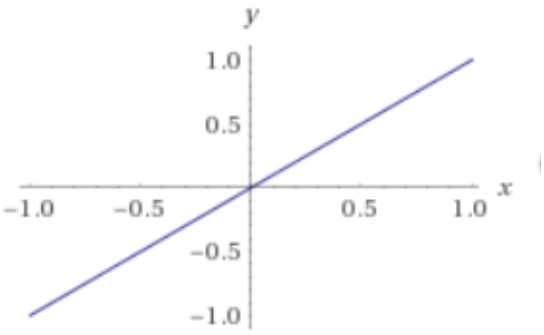
1. a point (x, y) lies on the graph if and only if the point $(-x, -y)$ lies on the graph.
2. a graph is symmetric about the origin if a rotation of 180° about the origin leaves the graph unchanged.
3. Notice that the definitions imply both x and $-x$ must be in the domain of f.

Example: $y = x^3, y = x, y = \sin x$.



(b)





Examples:

1. $y = x^2 + x$
2. $y = 2|t| + 1$
3. $y = \frac{x}{x^2-1}$
4. $y = \sin 2x$

Common Functions.

Linear Functions. $y = mx + b$, for constants m and b.

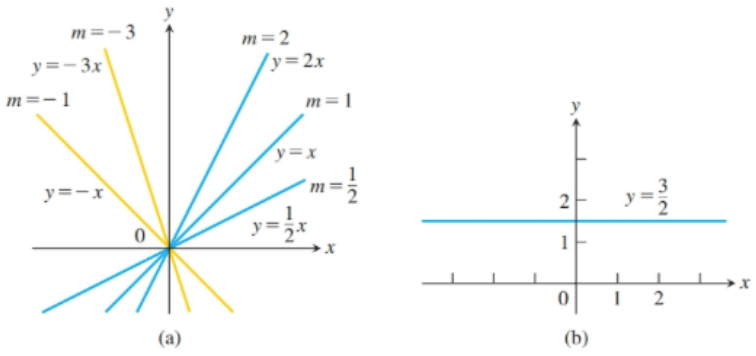


FIGURE 1.14 (a) Lines through the origin with slope m. (b) A constant function with slope $m = 0$.

Power Functions. $f(x) = x^a$, where a is a constant.

(a) $a = n$, a positive integer.

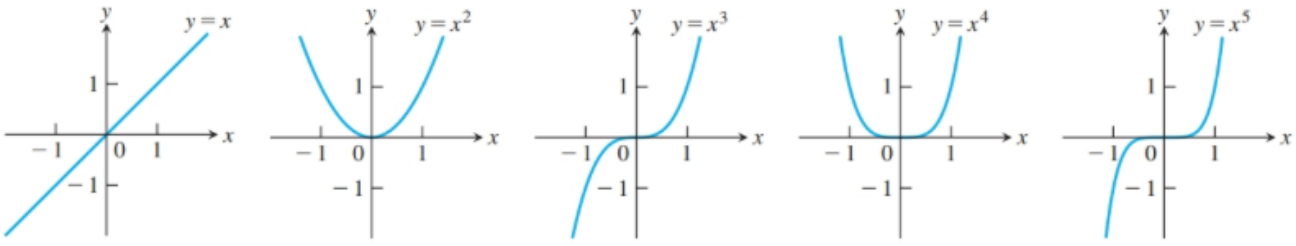
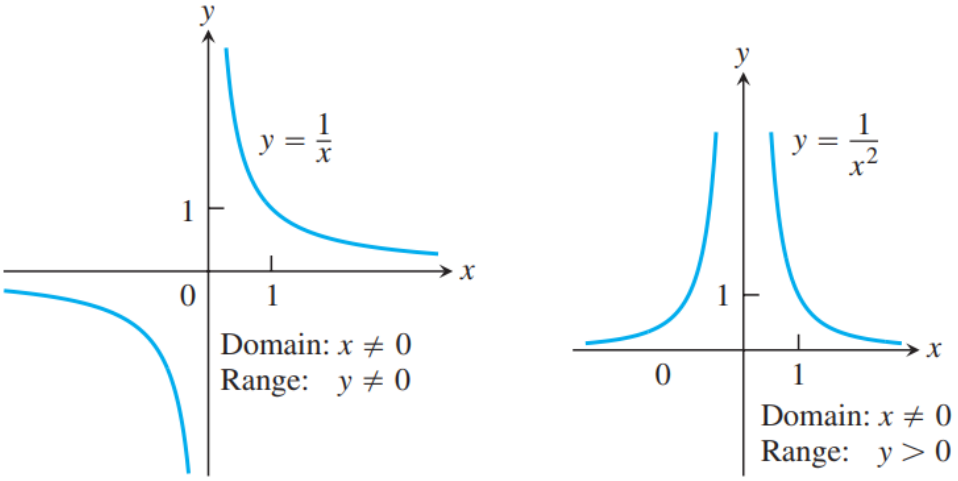


FIGURE 1.15 Graphs of $f(x) = x^n, n = 1, 2, 3, 4, 5$, defined for $-\infty < x < \infty$.

(b) $a = -1, a = -2$.

The graph of the functions $f(x) = x^{-1} = \frac{1}{x}$ and $g(x) = x^{-2} = \frac{1}{x^2}$, where $x \neq 0$.



(c) $a = \frac{1}{2}, \frac{1}{3}, \frac{3}{2}, \frac{2}{3}$

We note $f(x) = x^{\frac{1}{2}} = \sqrt{x}$ (square root) and $g(x) = x^{\frac{1}{3}} = \sqrt[3]{x}$ (cube root) with domain $[0, \infty)$, but $g(x) = x^{\frac{1}{3}} = \sqrt[3]{x}$ is defined for all real x. Recall that $x^{\frac{3}{2}} = \left(x^{\frac{1}{2}}\right)^3$ and $x^{\frac{2}{3}} = \left(x^{\frac{1}{3}}\right)^2$.

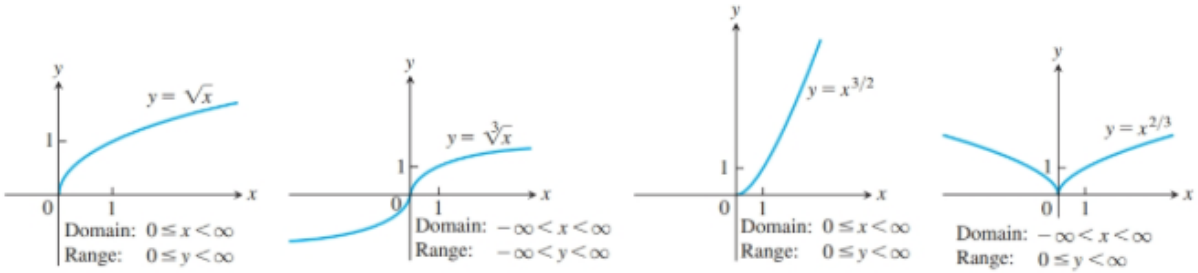


FIGURE 1.17 Graphs of the power functions $f(x) = x^a$ for $a = \frac{1}{2}, \frac{1}{3}, \frac{3}{2}$, and $\frac{2}{3}$.

Polynomials A function $p(x) = a_n x^n + a_{n-1} x^{n-1} + \cdots + a_1 x + a_0$, where n is a non-negative integer and the numbers $a_0, a_1, a_2, \ldots, a_n$ are real constants (**coefficients** of the polynomial).

Note:

1. All polynomials have domain $(-\infty, \infty)$.
2. If the leading coefficient $a_n \neq 0$ and $n > 0$, then n is called the **degree** of the polynomial.

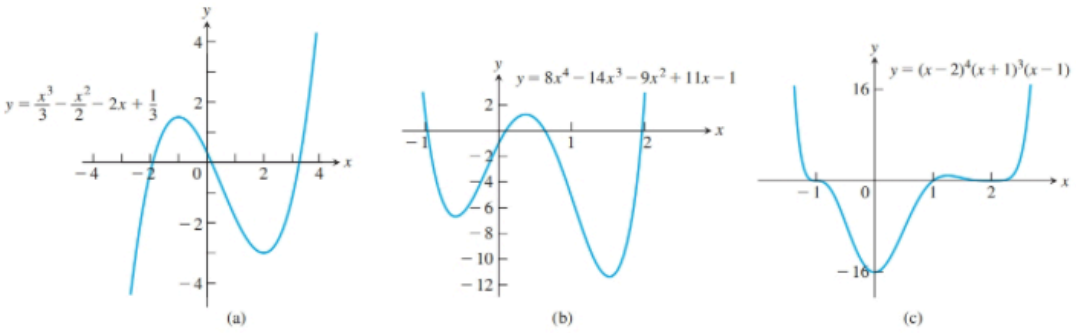
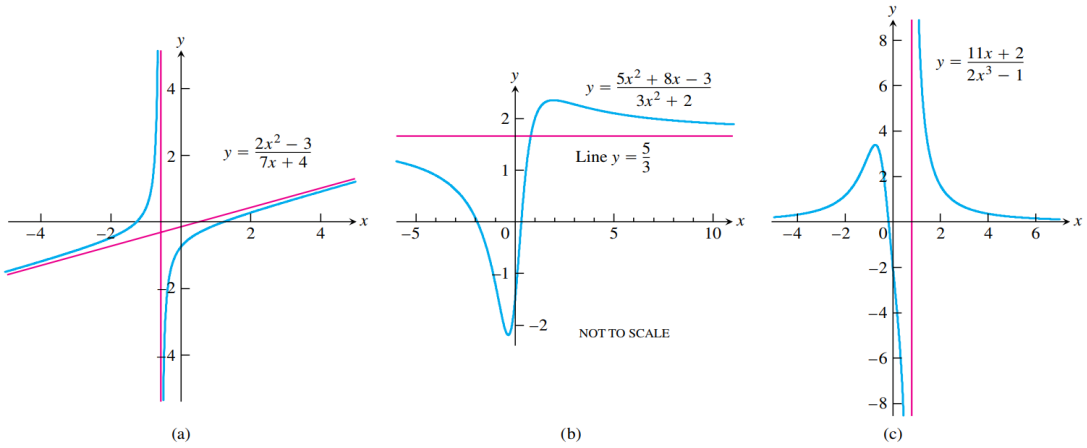


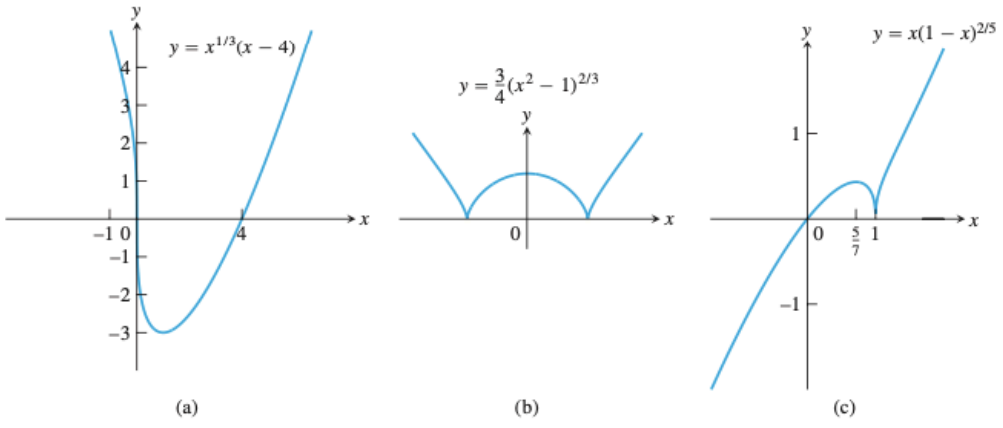
FIGURE 1.18 Graphs of three polynomial functions.

Rational Function. $f(x) = \frac{p(x)}{q(x)}$, where p, q are polynomials. The domain of a rational function is the set of all real x for which $q(x) \neq 0$.

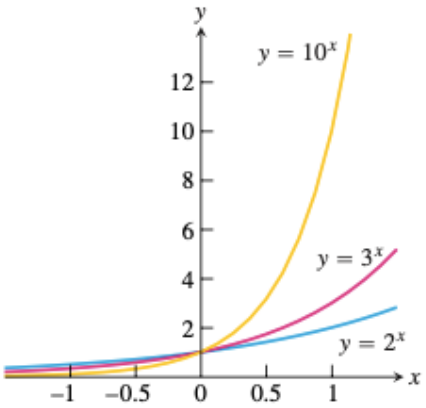
Example. $f(x) = \frac{2x^2-3}{7x+4}, x \neq -\frac{4}{7}$. Domain of f is $\left\{ x \mid x \neq -\frac{4}{7} \right\}$.



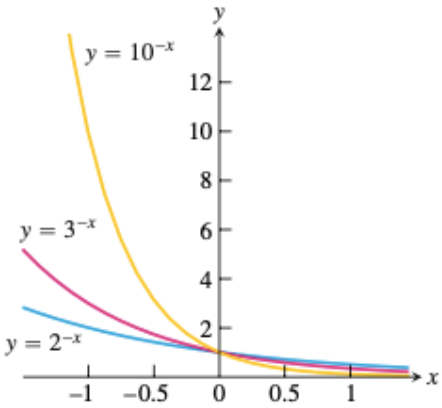
Algebraic functions Any function constructed from polynomials using the operations of algebra (addition, subtraction, multiplication, division, and roots) on polynomials or rational functions.



Exponential functions $f(x) = a^x$, where base $a \neq 1$ is a positive real number. The domain is **R** and range $(0, \infty)$.
Note: Exponential function never assumes the value 0.

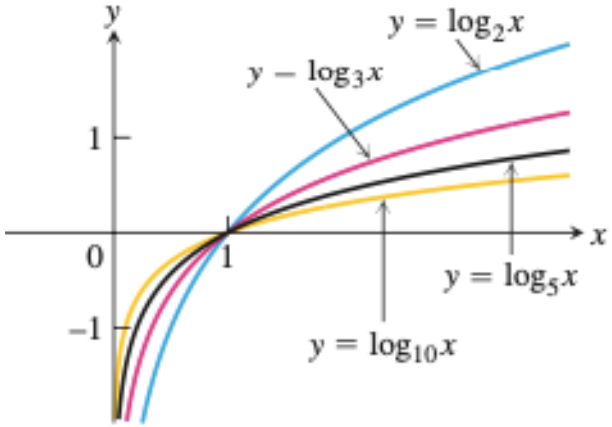


(a) $y = 2^x, y = 3^x, y = 10^x$



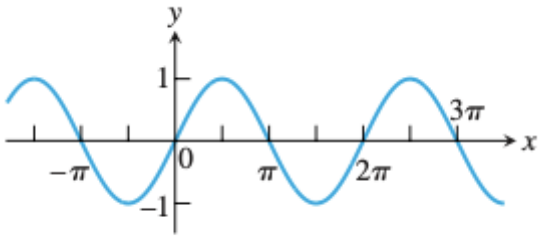
(b) $y = 2^{-x}, y = 3^{-x}, y = 10^{-x}$

Logarithmic functions $f(x) = \log_a x$, where the base $a > 0, a \neq 0$. The domain is $(0, \infty)$.
Note: They are inverse functions of the exponential functions.

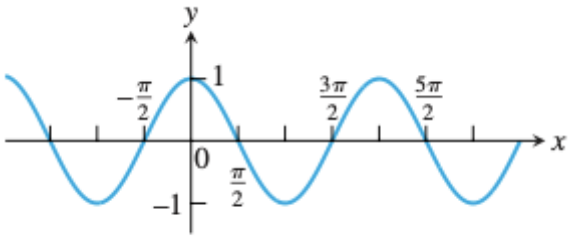


Trigonometric functions

Example. sin *x*, cos *x*, tan *x*, cot *x*, sec *x*, and cosec *x*.



(a) $f(x) = \sin x$



(b) $f(x) = \cos x$

Transcendental Functions These are functions that are not algebraic. They include the trigonometric, inverse trigonometric, exponential, and logarithmic functions, and many other functions as well.

Examples:

1. Does $\lceil -x \rceil = -\lfloor x \rfloor$ for all real x?

Combining Functions; Shifting and Scaling Graphs (pg 14)

Sums, Differences, Products, and Quotients

If f and g are functions, then for every x that belongs to the domains of both f and g (that is, for $x \in D(f) \cap D(g)$), we define functions $f + g$, $f - g$ and fg by the formulas

$$(f + g)(x) = f(x) + g(x)$$
$$(f - g)(x) = f(x) - g(x)$$
$$(fg)(x) = f(x)g(x).$$

At any point of $D(f) \cap D(g)$ at which $g(x) \neq 0$, we can also define the function f/g by the formula

$$\left(\frac{f}{g}\right)(x) = \frac{f(x)}{g(x)} \text{ (where } g(x) \neq 0).$$

Functions can also be multiplied by constants: If c is a real number, then the function cf is defined for all x in the domain of f by

$$(cf)(x) = cf(x).$$

Example: The functions defined by the formulas

$$f(x) = \sqrt{x} \text{ and } g(x) = \sqrt{1 - x}$$

have domains $D(f) = [0, \infty)$ and $D(g) = (-\infty, 1]$.

Function	Formula	Domain
$f + g$	$(f + g)(x) = \sqrt{x} + \sqrt{1 - x}$	$[0, 1] = D(f) \cap D(g)$
$f - g$	$(f - g)(x) = \sqrt{x} - \sqrt{1 - x}$	$[0, 1]$
$g - f$	$(g - f)(x) = \sqrt{1 - x} - \sqrt{x}$	$[0, 1]$
$f \cdot g$	$(f \cdot g)(x) = f(x)g(x) = \sqrt{x(1 - x)}$	$[0, 1]$
f/g	$\frac{f}{g}(x) = \frac{f(x)}{g(x)} = \sqrt{\frac{x}{1 - x}}$	$[0, 1)$ ($x = 1$ excluded)
g/f	$\frac{g}{f}(x) = \frac{g(x)}{f(x)} = \sqrt{\frac{1 - x}{x}}$	$(0, 1]$ ($x = 0$ excluded)

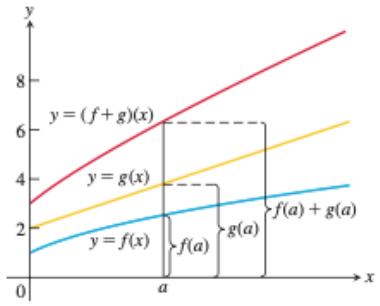


FIGURE 1.25 Graphical addition of two functions.

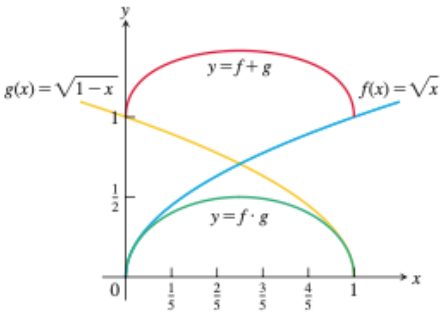


FIGURE 1.26 The domain of the function $f + g$ is the intersection of the domains of f and g, the interval $[0, 1]$ on the x-axis where these domains overlap. This interval is also the domain of the function $f \cdot g$ (Example 1).

Composing Functions

Definition.
If f and g are functions., the composite function $f \circ g$ ("f composed with g") is defined by

$$(f \circ g)(x) = f\big(g(x)\big).$$

The domain of $f \circ g$ consists of the numbers x in the domain of g for which $g(x)$ lies in the domain of f.

Note:

1. $f \circ g$ can be formed when the range of g lies in the domain of f.
2. Find $(f \circ g)(x)$, first find $g(x)$ and second find $f\big(g(x)\big)$.

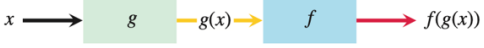


FIGURE 1.27 A composite function $f \circ g$ uses the output $g(x)$ of the first function g as the input for the second function f.

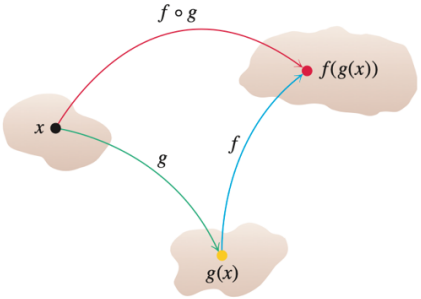


FIGURE 1.28 Arrow diagram for $f \circ g$. If x lies in the domain of g and $g(x)$ lies in the domain of f, then the functions f and g can be composed to form $(f \circ g)(x)$.

Example: The function $y = \sqrt{1 - x^2}$. First, we calculate $1 - x^2$ and then taking the square root of the result. The function y is the composite of the function $g(x) = 1 - x^2$ and function $f(x) = \sqrt{x}$.
Note that $1 - x^2$ cannot be negative (range of g). The domain of the composite is $[-1,1]$.

Example: If $f(x) = \sqrt{x}$ and $g(x) = x + 1$.

a. $(f \circ g)(x) = f\big(g(x)\big) = \sqrt{g(x)} = \sqrt{x+1}$. Domain: $[-1, \infty)$
b. $(g \circ f)(x) = g\big(f(x)\big) = f(x) + 1 = \sqrt{x} + 1$. Domain: $[0, \infty)$
c. $(f \circ f)(x) = f\big(f(x)\big) = \sqrt{f(x)} = \sqrt{\sqrt{x}} = x^{\frac{1}{4}}$. Domain: $[0, \infty)$
d. $(g \circ g)(x) = g\big(g(x)\big) = g(x) + 1 = (x+1) + 1 = x + 2$. Domain: $(-\infty, \infty)$

Shifting a Graph of a Function (pg 16)

Shifting Formulas

Vertical Shifts: $y = f(x) + k$ Shifts the graph of f **up** k units if $k > 0$
Shifts the **down** $|k|$ units if $k < 0$

Horizontal Shifts: $y = f(x + h)$ Shifts the graph of f **left** h units if $h > 0$
Shifts the **right** $|h|$ units if $h < 0$

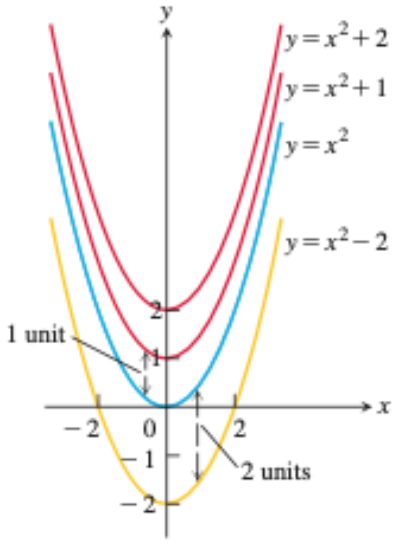


FIGURE 1.29 To shift the graph of $f(x) = x^2$ up (or down), we add positive (or negative) constants to the formula for f (Examples 3a and b).

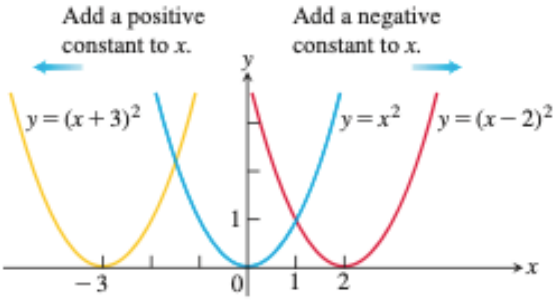


FIGURE 1.30 To shift the graph of $y = x^2$ to the left, we add a positive constant to x (Example 3c). To shift the graph to the right, we add a negative constant to x.

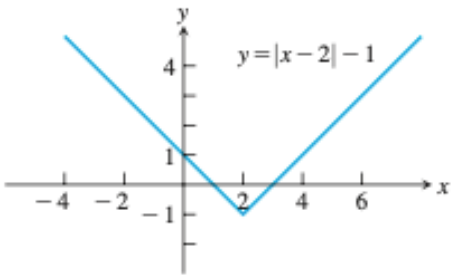


FIGURE 1.31 The graph of $y = |x|$ shifted 2 units to the right and 1 unit down (Example 3d).

Vertical and Horizontal Scaling and Reflecting Formulas

For $c > 1$, the graph is scaled:

$y = cf(x)$ Stretches the graph of f vertically by a factor of c.

$y = \frac{1}{c}f(x)$ Compresses the graph of f vertically by a factor of c.

$y = f(cx)$ Compresses the graph of f horizontally by a factor of c.

$y = f\left(\frac{x}{c}\right)$ Stretches the graph of f horizontally by a factor of c.

For $c = -1$, the graph is reflected:

$y = -f(x)$ Reflects the graph of f across the x-axis.

$y = f(-x)$ Reflects the graph of f across the y-axis.

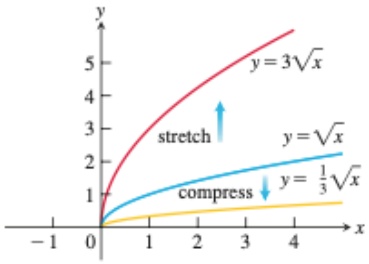


FIGURE 1.32 Vertically stretching and compressing the graph $y = \sqrt{x}$ by a factor of 3 (Example 4a).

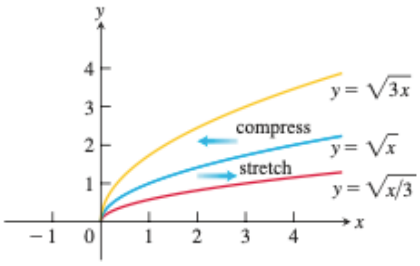


FIGURE 1.33 Horizontally stretching and compressing the graph $y = \sqrt{x}$ by a factor of 3 (Example 4b).

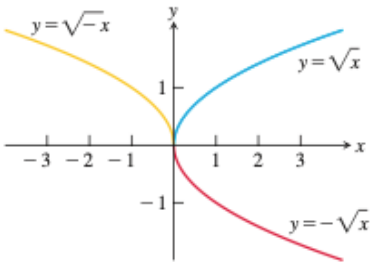


FIGURE 1.34 Reflections of the graph $y = \sqrt{x}$ across the coordinate axes (Example 4c).

EXAMPLE 5 Given the function $f(x) = x^4 - 4x^3 + 10$ (Figure 1.35a), find formulas to

(a) compress the graph horizontally by a factor of 2 followed by a reflection across the y-axis (Figure 1.35b).

(b) compress the graph vertically by a factor of 2 followed by a reflection across the x-axis (Figure 1.35c).

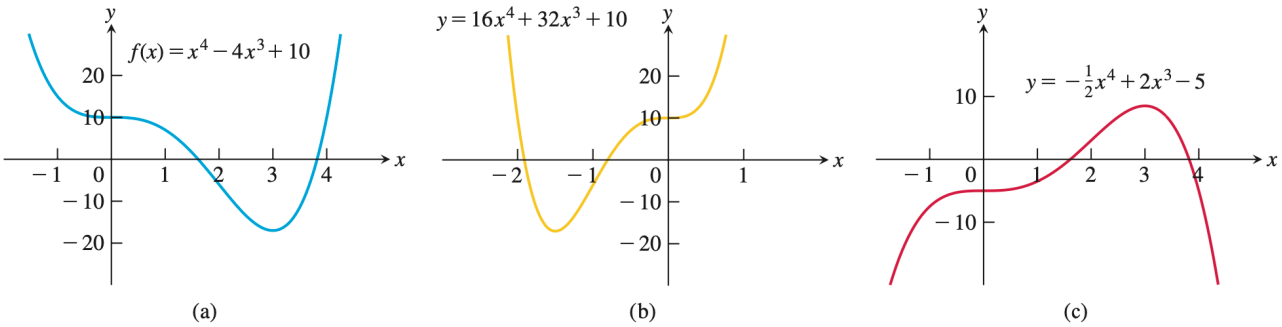


FIGURE 1.35 (a) The original graph of f. (b) The horizontal compression of $y = f(x)$ in part (a) by a factor of 2, followed by a reflection across the y-axis. (c) The vertical compression of $y = f(x)$ in part (a) by a factor of 2, followed by a reflection across the x-axis (Example 5).

Example:

1. Graph the functions in
   (a) $y = \sqrt{9 - x}$, → $\sqrt{-(x-9)}$
   (b) $y = |1 - x| - 1$.

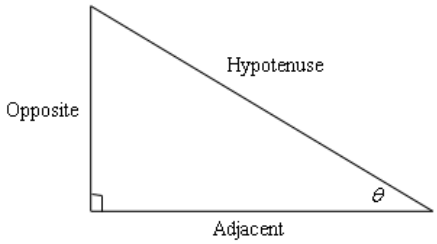
Trigonometry Functions

Radius Measure
In calculus, it is best to see units called **radians**, because of the way they simplify later calculation.
Note: π radians = 180 degrees.

The six basic trigonometric functions are

sine: $\sin\theta$	**cosecant**: $cosec\ \theta$
cosine: $\cos\theta$	**secant**: $\sec\theta$
tangent: $\tan\theta$	**cotangent**: $\cot\theta$

All trigonometric functions can be defined in terms of the sides of a right triangle:



$$\cos\theta = \frac{\text{adjacent}}{\text{hypotenuse}}, \qquad \tan\theta = \frac{\text{opposite}}{\text{adjacent}}, \qquad \sin\theta = \frac{\text{opposite}}{\text{hypotenuse}}.$$

TABLE 1.1 Angles measured in degrees and radians

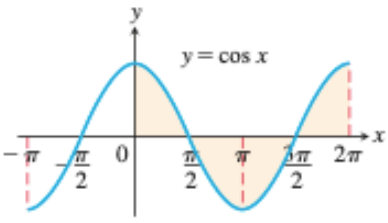
Degrees	-180	-135	-90	-45	0	30	45	60	90	120	135	150	180	270	360
θ (radians)	$-\pi$	$\frac{-3\pi}{4}$	$\frac{-\pi}{2}$	$\frac{-\pi}{4}$	0	$\frac{\pi}{6}$	$\frac{\pi}{4}$	$\frac{\pi}{3}$	$\frac{\pi}{2}$	$\frac{2\pi}{3}$	$\frac{3\pi}{4}$	$\frac{5\pi}{6}$	π	$\frac{3\pi}{2}$	2π

Example. Evaluate $\sin\left(\frac{2\pi}{3}\right)$ and $\sin\left(-\frac{2\pi}{3}\right)$ without using the calculator.

We have $\sin(2\pi/3) = \sqrt{3}/2$. The angle $-2\pi/3$ can be reached by rotating an angle of $2\pi/3$ in the clockwise direction from the positive x-axis. In doing so, we will reach the angle $4\pi/3$. Therefore, $\sin(-2\pi/3) = \sin(4\pi/3) = -\sqrt{3}/2$. Notice that $\sin(-2\pi/3) = -\sin(2\pi/3)$.

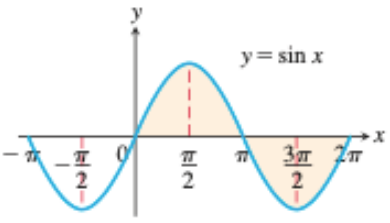
Periodicity and Graphs of the Trigonometric Functions

Definition: A function $f(x)$ is **periodic** if there is a positive number p such that $f(x+p) = f(x)$ or $f(x-p) = f(x)$ for every value of x. The smallest such value of p is the **period** of f.



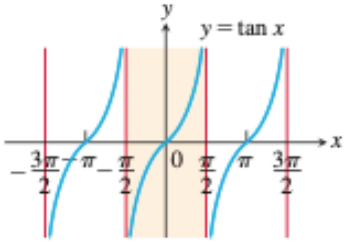
Domain: $-\infty < x < \infty$
Range: $-1 \le y \le 1$
Period: 2π

(a)



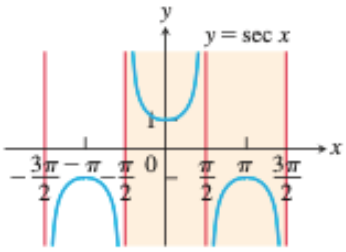
Domain: $-\infty < x < \infty$
Range: $-1 \le y \le 1$
Period: 2π

(b)



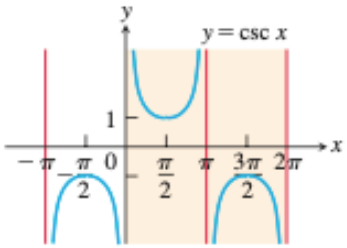
Domain: $x \neq \pm\frac{\pi}{2}, \pm\frac{3\pi}{2}, \ldots$
Range: $-\infty < y < \infty$
Period: π

(c)



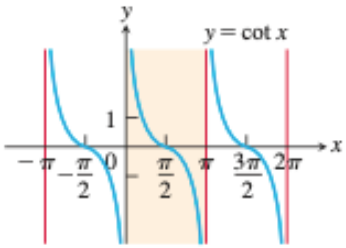
Domain: $x \neq \pm\frac{\pi}{2}, \pm\frac{3\pi}{2}, \ldots$
Range: $y \le -1$ or $y \ge 1$
Period: 2π

(d)



Domain: $x \neq 0, \pm\pi, \pm 2\pi, \ldots$
Range: $y \le -1$ or $y \ge 1$
Period: 2π

(e)



Domain: $x \neq 0, \pm\pi, \pm 2\pi, \ldots$
Range: $-\infty < y < \infty$
Period: π

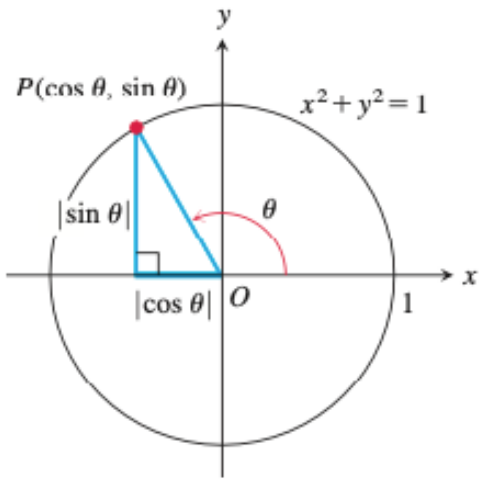
(f)

Example:
What is the period of each function?

1. $y = \sin 2x$
2. $y = \sin\left(\frac{x}{2}\right)$
3. $y = \cos\left(\frac{\pi x}{2}\right)$
4. $y = \sin\left(x + \frac{\pi}{6}\right)$

Even:
$\cos(-x) = \cos x$

Odd:
$\sin(-x) = -\sin x$
$\tan(-x) = -\tan x$



The coordinates of any point $P(x, y)$ in the plane can be **expressed in terms of the point's distance r from the origin** and the angle θ that ray OP makes with the positive x-axis.

Since $\frac{x}{r} = \cos\theta, \frac{y}{r} = \sin\theta$.

Hence, $x = r\cos\theta\,, y = r\sin\theta$.

Trigonometric Identities

$$\cos^2\theta + \sin^2\theta = 1$$

$$1 + \tan^2\theta = \sec^2\theta$$

$$1 + \cot^2\theta = \csc^2\theta$$

Addition Formulas

$$\cos(A + B) = \cos A\cos B - \sin A\sin B$$

$$\sin(A + B) = \sin A\cos B + \cos A\sin B$$

Double-Angle Formulas

$$\cos(2\theta) = \cos^2\theta - \sin^2\theta$$

$$\sin(2\theta) = 2\sin\theta\cos\theta$$

Half-Angle Formulas

$$\cos^2(\theta) = \frac{1 + \cos 2\theta}{2}$$

$$\sin^2(\theta) = \frac{1 - \cos 2\theta}{2}$$

The Law of Cosines

$$c^2 = a^2 + b^2 - 2ab\cos\theta$$

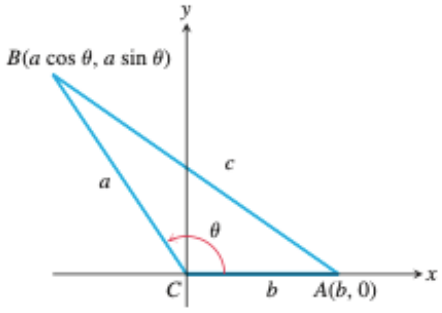


FIGURE 1.46 The square of the distance between A and B gives the law of cosines.

Two Special Inequalities

$$-|\theta| \le \sin\theta \le |\theta| \text{ and } -|\theta| \le 1 - \cos\theta \le |\theta|$$

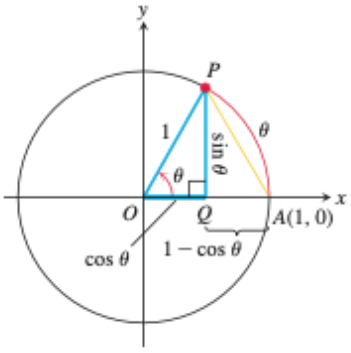


FIGURE 1.47 From the geometry of this figure, drawn for $\theta > 0$, we get the inequality $\sin^2\theta + (1 - \cos\theta)^2 \le \theta^2$.

Examples (exercise 1.3):

1. Solve $\cos 2\theta + \cos\theta = 0$ for $0 \le \theta \le 2\pi$.
2. Evaluate $\cos\left(\frac{\pi}{12}\right)$.